LESSON 9.1

Explore Relationships Between Two Variables

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

REMEMBER

Explore Relationships Between Two Variables

Independent & Dependent Variables When two quantities are related, one of them depends on the other. Naming which is which is the whole job of this lesson.

1. Independent Variable (x): The input / cause that is chosen or changes freely.
2. Dependent Variable (y): The output / effect that depends on the independent variable.
3. Sentence check: 'The (dependent) depends on the (independent)'

Words: independent variable *(variable independiente)* · dependent variable *(variable dependiente)* · depends on *(depende de)* · quantity *(cantidad)* · relationship *(relación)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each quantity: independent or dependent?

Groups: Independent variable (input) · Dependent variable (output)

_____ Hours of sunlight a tray of seedlings gets

_____ Months of gym membership

_____ Minutes spent jogging

_____ Height the seedlings grow

_____ Total cost of the membership

_____ Calories burned on the jog

HINT Use the sentence check: 'The ___ depends on the ___.' The first blank is dependent.

2 **Complete the table. Show how you found each value.**

Claire burns Calories at a steady rate: 30 minutes burns 123 Calories. Complete the table.

Minutes jogged (independent)	Thinking	Calories burned (dependent)
30		
60		
90		

HINT 60 minutes is twice 30 minutes, so it burns twice the Calories.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

If Claire runs exactly 3 miles and records how long it takes, which is the independent variable now?

(A) Time it takes
(B) Her speed
(C) Distance (3 miles)
(D) Calories burned

HINT What did Claire decide in advance this time?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A ticket costs $8. Which equation represents "8 tickets cost $56"?

(A) $8t = 56$
(B) $t + 8 = 56$
(C) $t - 8 = 56$
(D) $t \div 8 = 56$

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Solve: 4x = 32

(A) x = 8

(B) x = 28

(C) x = 36

(D) x = 128

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Maya needs at least 20 points to win a prize. Which inequality shows this?

(A) $p \geq 20$

(B) $p > 20$

(C) $p \leq 20$

(D) $p < 20$

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you and your partner put in different columns. Say the sentence 'The ____ depends on the ____' out loud both ways — which version is true, and how does that settle which variable is independent?

SENTENCE STARTER The ___ depends on the ___, so ___ is the independent variable.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS Akela planted three trays of bean seeds for a science experiment. Each tray sat in a different spot, so each got a different amount of sunlight per day. After three weeks she measured the average plant height in every tray. Two quantities in that experiment are related: change the sunlight and the height changes with it. One of them Akela chose, and the other one answered back. Today you will name which quantity is the independent one and which depends on it, and you will find pairs like this in a jogging tracker and a pizza shop.

1 **Circle the letter of the best answer. Show how you know.**

Solve: x + 7 = 19

(A) x = 7
(B) x = 12
(C) x = 19
(D) x = 26

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

At a local pizza shop a customer can order up to three toppings on a large pizza. The shop has 6 employees. The shop spends $1.15 on cheese and $1.95 on sauce for each pizza it makes. The number of pizzas made each week determines other costs at the shop. Which variable is the independent variable in this relationship?

(A) the cost of cheese used
(B) the number of pizzas made
(C) the number of employees
(D) the number of toppings allowed

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

At a local pizza shop a customer can order up to three toppings on a large pizza. The shop has 6 employees. The shop spends $1.15 on cheese and $1.95 on sauce for each pizza it makes. Which two quantities have a dependent relationship in this scenario?

(A) number of employees and number of pizzas made

(B) cost of cheese and number of pizzas made

(C) number of toppings allowed and cost of sauce

(D) number of employees and number of toppings allowed

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

At a local pizza shop a customer can order up to three toppings on a large pizza. The shop has 6 employees. The shop spends $1.15 on cheese and $1.95 on sauce for each pizza it makes. The shop makes 20 pizzas in a week. Which of these does NOT depend on the number of pizzas sold?

(A) the total cost of cheese

(B) the number of employees

(C) the total cost of sauce

(D) the total spent on cheese and sauce combined

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A dog walker charges by the hour. Which sentence is correct?

(A) The number of hours depends on the total charge, so the charge is the independent variable.

(B) Both quantities are independent variables.

(C) The total charge depends on the number of hours, so hours is the independent variable.

(D) Neither quantity depends on the other.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a card you and your partner put in different columns. Say the sentence 'The ____ depends on the ____' out loud both ways — which version is true, and how does that settle which variable is independent?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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